

Mobile Application Development Lab

Experiment: Multi-Page Consumer App with Database Integration

Objective

To design and develop a multi-screen mobile application that demonstrates data transfer between screens, basic UI design, simple calculations, and local database storage and retrieval.

Tools/Technologies

- Android Studio (Java/Kotlin)
- iOS (Swift / SwiftUI / UIKit)

Part A: UI, Navigation and Logic

Screen 1: Consumer Details Page

Input Fields:

- Consumer Number
- First Name
- Last Name
- Mobile Number

Button: Next

On Click: Pass Consumer Number and First Name to Screen 2.

Screen 2: Welcome & Input Page

Display Message: Welcome "First Name"

Input Fields:

- Total Units Consumed
- Rate per Unit

Button: Calculate

On Click: Calculate Total Cost = Units $\times$ Rate and pass data to Screen 3.

Screen 3: Result Page

Display:

- Consumer Number
- Total Cost

UI Design Constraints

- Use Relative Layouts / ConstraintLayout / AutoLayout
- Each page must have different background color (use color names only)
- All text must have different colors (no hex codes)
- At least one page must contain an image

Part B: Database Integration

Screen 1: Registration Page

Input Fields:

- Consumer Number
- First Name
- Password

Button: Save

Store data in local database (SQLite).

Screen 2: Display Page

Retrieve and display:

- Consumer Number
- First Name
- Password

Evaluation Criteria

UI Design & Layout – 10 Marks

Navigation & Data Passing – 10 Marks

Logic Implementation – 10 Marks

Database Operations – 10 Marks

Code Quality – 10 Marks

Total – 50 Marks

Submission Requirements

- Source Code (ZIP / GitHub)
- APK / IPA (if possible)
- Screenshots of all screens
- Short documentation (1–2 pages)